Supplementary Appendix

This appendix has been provided by the authors to give readers additional information about their work.

Supplement to: Chi KN, Rathkopf DE, Smith MR, et al. Niraparib and abiraterone acetate for metastatic castration-resistant prostate cancer.

Supplementary Materials for

Niraparib and abiraterone for metastatic castration-resistant prostate cancer

Kim N. Chi, MD,[1] Dana E. Rathkopf, MD,[2] Matthew R. Smith, MD,[3] Eleni Efstathiou, MD,[4] Gerhardt Attard, MD,[5] David Olmos, MD,[6] Ji Youl Lee, MD,[7] Eric J. Small, MD,[8] Andrea J. Pereira de Santana Gomes, MD,[9] Guilhem Roubaud, MD,[10] Marniza Saad, MD,[11] Bogdan Zurawski, MD,[12] Valerii Sakalo, MD,[13] Gary E. Mason, MD,[14] Peter Francis, MD,[15] George Wang, MS, MAS,[14] Daphne Wu, PhD,[16] Brooke Diorio, PhD,[17] Angela Lopez-Gitlitz, MD,[16] Shahneen Sandhu, MD[18]; on behalf of the MAGNITUDE investigators

[1]University of British Columbia, BC Cancer – Vancouver Center, Vancouver, BC, Canada; [2]Memorial Sloan Kettering Cancer Center and Weill Cornell Medicine, New York, NY, USA; [3]Massachusetts General Hospital Cancer Center and Harvard Medical School, Boston, MA, USA; [4]Houston Methodist Cancer Center, Houston, TX, USA; [5]University College London, London, UK; [6]Department of Medical Oncology, Hospital Universitario 12 de Octubre, Instituto de Investigación Sanitaria Hospital 12 de Octubre (imas12), Madrid, Spain; [7]Department of Urology Cancer Center, Seoul St. Mary's Hospital, The Catholic University of Korea, Seoul, South Korea; [8]Helen Diller Family Comprehensive Cancer Center, University of California San Francisco, San Francisco, CA, USA; [9]Liga Norte Riograndense Contra o Câncer, Natal, Brazil; [10]Department of Medical Oncology, Institut Bergonié, Bordeaux, France; [11]Department of Clinical Oncology, Faculty of Medicine, University of Malaya, Kuala Lumpur, Malaysia; [12]Department of Outpatient Chemotherapy, Professor Franciszek Lukaszczyk Oncology Center, Bydgoszcz, Poland; [13]Kyiv City Clinical Oncology Center and Academician O.F.Vozianov Institute of Urology of the National Academy of Medical Sciences of Ukraine, Kyiv, Ukraine; [14]Janssen Research & Development, LLC, Spring House, PA, USA; [15]Janssen Research & Development, LLC, Raritan, NJ, USA; [16]Janssen Research & Development, LLC, Los Angeles, CA, USA; [17]Janssen Research & Development, LLC, Titusville, NJ, USA; [18]Peter MacCallum Cancer Center and the University of Melbourne, Melbourne, Australia.

Contents

Supplementary Methods

Additional Methods

Patients enrolled were stratified by prior taxane chemotherapy (yes vs no), prior androgen receptor–targeted therapy (yes vs no), and prior AAP use in the mCRPC setting (yes vs no). In addition, for the HRR+ cohort, stratification by gene alteration group (*BRCA1* or *BRCA2* vs all other gene alterations) was performed. Patient-level unblinding was permitted after determination of radiographic progression by central review at the request of the investigator and approved by the unblinding committee.

The overall approach for the efficacy analysis was as follows: rPFS in the *BRCA1/2* subgroup of the HRR+ cohort was tested first at a 2-sided alpha level of 0.05; if this met statistical significance, then this alpha would be passed to rPFS in the full HRR+ cohort, which would be tested at the 0.05 level; if rPFS for all HRR+ patients showed statistical significance, then the alpha would be passed to the secondary endpoints, which would be tested in the HRR+ cohort using a group sequential method. The Kaplan-Meier product limit method and a stratified Cox model were used to estimate the time-to-event variables and to obtain HRs and associated confidence intervals (CIs). Unless otherwise specified, stratified log-rank tests were used to test the treatment effect for time-to-event variables, and response rate variables were evaluated using the chi square statistic or the exact test if the cell counts were small.

Endpoints

Secondary endpoint definitions were as follows: time to initiation of cytotoxic chemotherapy, defined as the time from the date of randomization to the date of initiation of cytotoxic chemotherapy for prostate cancer; time to symptomatic progression, defined as the earliest time to any of the following: use of external beam radiation therapy for skeletal symptoms, the need for tumor-related orthopedic surgical intervention, other cancer-related procedures, cancer-related morbid events, or initiation of a new systemic anticancer therapy because of cancer pain; and overall survival, defined as the time from the date of randomization to the date of death from any cause. Additional endpoints were defined as follows: time to prostate-specific antigen progression (time from randomization to the first date of documented prostate-specific antigen progression per Prostate Cancer Working Group 3 [PCWG3] criteria[1]); objective response rate (the proportion of patients with measurable disease whose best response is either complete response or partial response by central review, as defined by Response Evaluation Criteria in Solid Tumors [RECIST] 1.1[2], with no evidence of bone progression according to the PCWG3 criteria[1]); time to pain progression (time from the date of randomization to an average increase of 2 points from baseline in the Brief Pain Inventory–Short Form [BPI-SF] worst pain intensity [Item 3] observed at 2 consecutive evaluations ≥3 weeks apart); duration of response (the time of documented response to the first date of documented disease progression in patients who had a measurable lesion at baseline according to RECIST 1.1[2] and had a tumor response of complete response or partial response post-baseline and before disease progression, as identified by RECIST 1.1[2]); prostate-specific antigen response rate (proportion of patients achieving a prostate-specific antigen decline of ≥50% and confirmed at 3–4 weeks later according to PCWG3 criteria by Week 12 and during the treatment period); and progression-free survival on first subsequent therapy (time from randomization to the date of progression [radiographic, clinical, or prostate-specific antigen progression] on the first subsequent therapy or death from any cause, whichever occurs first).

Censoring Rules

For the primary endpoint of radiographic progression-free survival, patients without radiographic progression or death were to be censored at the last disease assessment date if never started subsequent anti-cancer therapy or censored at the last disease assessment date prior to the start of the subsequent anti-cancer therapy if started subsequent anticancer therapy. Key censoring rules are summarized below.

Scenario	**Censoring rule**
No disease assessment at baseline or No disease assessment after baseline	Censored on the date of randomization
Patients who are lost to follow-up or withdraw from study	Censored on the date of the last disease assessment

Patients who receive new systemic anti-cancer therapy known or intended for the treatment of mCRPC during the study prior to documented disease progression or death	Censored on the date of the last disease assessment prior to the start of the new systemic anti-cancer therapy
Patients with no evidence of radiographic progressive disease or death	Censored on the date of the last disease assessment
Patients who miss ≥2 consecutive planned radiographic scans or has ≥2 consecutive unevaluable scans before progression or death	Censored on the date of the last disease assessment before the missed/unevaluable scans
No postbaseline assessment and death occurred after missed 2 or more planned disease assessments	Censored on the date of randomization

In terms of secondary endpoints, for overall survival, patients alive at time of analysis would be censored on the last date the patient was known to be alive. For both time to initiation of cytotoxic chemotherapy and time to symptomatic progression, patients who did not have event observed at the time of the analysis were to be censored on the last known alive date. For time to pain progression, patients with no pain progression at the time of analysis were to be censored at the last date of BPI-SF pain assessment. For all these time-to-events endpoints, patients with no on-study assessment or no baseline assessment were to be censored on the date of randomization.

Cohort 2 Futility Analysis

A futility analysis was planned to be conducted for the homologous recombination repair (HRR)–negative cohort after approximately 200 patients were enrolled and approximately 125 composite progression events had occurred. The HRR- cohort was considered futile if the observed hazard ratio for composite radiographic progression-free survival or prostate-specific antigen progression events (whichever occurred first) was greater than or equal to 1. Based on results of the futility analysis, the Independent Data Monitoring Committee (IDMC) was to recommend whether the study would continue enrollment to the planned sample size of approximately 600 patients or discontinue enrollment in this cohort. If futility was declared, the HRR+ and HRR- cohorts would be assessed separately. Prior to the analysis, a protocol amendment occurred to include patients with *CDK12* alterations in the HRR+ cohort instead of the HRR- cohort. This inclusion of *CDK12* and broadening of the biomarker-selected population was based upon data from the PROfound study and US Food and Drug Administration approval of olaparib in a broad panel of genes including *CDK12*.[3,4] New patients with *CDK12* alterations were from then onwards enrolled into the HRR+ cohort and were to be included in the HRR+ analysis. A total of 14 patients with *CDK12* alterations who were previously enrolled into the HRR- cohort remained blinded and were to be excluded from the futility analysis (events from these patients were not counted as composite progression events); these patients were to be included in the sensitivity analysis in the HRR+ cohort.

Patient-reported Outcomes

Patient-reported outcomes collected were to evaluate patient experience in terms of pain, prostate cancer symptoms, treatment-related tolerability, and overall health-related QoL. Disease-related symptoms were to be assessed using the BPI-SF and the prostate cancer symptom subscale of the Functional Assessment of Cancer Therapy–Prostate (FACT-P) questionnaire. Overall health-related QoL was to be measured with the FACT-P and the EQ-5D-5L to assess any impact from either disease-related or treatment-related experiences. Patient-reported outcomes were to be collected per the protocol schedule. All visit-specific assessments were to be conducted and completed before any tests, procedures, or other consultations for that visit to prevent influencing patient perceptions. Post-progression data were also to be collected to understand the patient experience beyond progression. Changes from baseline were compared between treatment arms using repeated measures analysis. FACT-P assessments were conducted on Day 1 of specified cycles, and threshold for deterioration was defined as ≥10 decrease in FACT-P total scale.

Trial Oversight

The study was performed in accordance with the protocol, principles of the Declaration of Helsinki, Good Clinical Practice guidelines, and applicable regulatory and country-specific requirements. An IDMC was commissioned to monitor data, ensure safety of patients enrolled in the study, and review efficacy information. The IDMC was composed of members not associated with the conduct of the study, except in their role on the IDMC. An independent biostatistician not affiliated with the project was assigned to prepare and provide study data to the IDMC. Complete details regarding the composition and governance of the IDMC is outlined in the IDMC charter.

The trial was designed, interpreted, and reported as a collaboration between the lead investigators and employees of Janssen Research & Development, LLC (the sponsor). Data were analyzed by the sponsor and provided confidentially to the authors. All the authors had full access to all the trial data, made the decision to submit the manuscript for publication, and vouch for the fidelity of the trial to the protocol. All the authors critically reviewed and approved the manuscript; medical writing and editing assistance was funded by Janssen Research & Development, LLC.

Supplementary Results

Supplementary Table S1. Baseline Characteristics of the HRR- Cohort

	Niraparib+AAP (n=123)	**Placebo+AAP (n=124)**	**Total (N=247)**
Median age (range), years	72.0 (53–87)	71.0 (52–85)	71.2 (52–87)
Mean body weight (SD), kg	80.3 (14.8)	82.5 (16.9)	81.4 (15.9)
Race, n (%)			
White	72 (58.5)	83 (66.9)	155 (62.8)
Asian	45 (36.6)	37 (29.8)	82 (33.2)
Native Hawaiian or other	1 (0.8)	0	1 (0.4)
Unknown	5 (4.1)	4 (3.2)	9 (3.6)
Median hemoglobin at baseline (range), g/L	128.0 (85.0–164.0)	129.0 (63.0–155.0)	129.0 (63.0–164.0)
Median lactate dehydrogenase at baseline (range), enzyme U/L[a]	194.0 (84.0–645.0)	202.0 (131.0–758.0)	199.0 (84.0–758.0)
Median prostate-specific antigen at baseline (range), µg/liter	26.6 (0.0–7,684.0)	20.6 (0.0–4,337.0)	22.3 (0.0–7684.0)
ECOG PS score, n (%)			
0	80 (65.0)	86 (69.4)	166 (67.2)
1	43 (35.0)	38 (30.6)	81 (32.8)
Extent of disease progression at study entry, n (%)			

Bone	112 (91.1)	106 (85.5)	218 (88.3)
Lymph node	45 (36.6)	60 (48.4)	105 (42.5)
Visceral	22 (17.9)	18 (14.5)	40 (16.2)
Adrenal gland	2 (1.6)	1 (0.8)	3 (1.2)
Liver	2 (1.6)	6 (4.8)	8 (3.2)
Lung	17 (13.8)	8 (6.5)	25 (10.1)
Other soft tissue	5 (4.1)	7 (5.6)	12 (4.9)
Metastasis stage at diagnosis, n (%)			
M0	36 (29.3)	48 (38.7)	84 (34.0)
M1	75 (61.0)	63 (50.8)	138 (55.9)
Unknown	12 (9.8)	13 (10.5)	25 (10.1)
Gleason score at diagnosis, n (%)			
<8	31 (25.2)	39 (31.5)	70 (28.3)
≥8	88 (71.5)	80 (64.5)	168 (68.0)
Unknown	4 (3.3)	5 (4.0)	9 (3.6)
Mean BPI-SF pain score, Item 3 (SD)	1.07 (1.46)	0.95 (1.42)	1.01 (1.44)
Prior therapies for prostate cancer, n (%)			

ADT[b]	120 (97.6)	123 (99.2)	243 (98.4)
Radiotherapy	51 (41.5)	56 (45.2)	107 (43.3)
Prostatectomy	63 (51.2)	68 (54.8)	131 (53.0)
AR-targeted therapy for nmCRPC or mCSPC	1 (0.8)	1 (0.8)	2 (0.8)
Taxane chemotherapy for mCSPC	23 (18.7)	21 (16.9)	44 (17.8)
AAP (≤4 months) for mCRPC	22 (17.9)	19 (15.3)	41 (16.6)
Other	23 (18.7)	23 (18.5)	46 (18.6)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; SD, standard deviation; ECOG PS, Eastern Cooperative Oncology Group performance status; BPI-SF, Brief Pain Inventory–Short Form; ADT, androgen deprivation therapy; AR, androgen receptor; nmCRPC, nonmetastatic castration-resistant prostate cancer; mCRPC, metastatic castration-resistant prostate cancer; mCSPC, metastatic castration-sensitive prostate cancer.

[a]N=245 for total population.

[b]Patients were permitted to undergo a bilateral orchiectomy instead of androgen deprivation therapy with a gonadotropin-releasing hormone agonist during the study.

Note: Percentages may not add up to 100% due to rounding.

Supplementary Table S2. Radiographic Progression-free Survival Multivariate Analysis

Model parameter	Model fit		Hazard ratio	
	Coeff. (SE)	***P* value**	**Coeff. (SE)**	**95% CI**
Treatment (niraparib vs placebo)	–0.427 (0.140)	0.0022	0.652	(0.496, 0.858)
PSA	0.084 (0.037)	0.0226	1.088	(1.012, 1.169)
Lactate dehydrogenase	0.370 (0.145)	0.0107	1.447	(1.090, 1.923)
Alkaline phosphatase	0.199 (0.098)	0.0412	1.221	(1.008, 1.478)
Presence of visceral disease (yes vs no)	0.475 (0.162)	0.0034	1.608	(1.171, 2.210)

Coeff., coefficient; SE, standard error; CI, confidence interval; PSA, prostate-specific antigen.

Supplementary Table S3. Overall Survival Multivariate Analysis

Model parameter	Model fit		Hazard ratio	
	Coeff. (SE)	*P* value	Coeff. (SE)	95% CI
Treatment (niraparib vs placebo)	–0.266 (0.193)	0.1682	0.767	(0.525, 1.119)
PSA	0.155 (0.051)	0.0021	1.168	(1.058, 1.290)
Lactate dehydrogenase	0.562 (0.160)	0.0004	1.755	(1.282, 2.402)
Average pain score (BPI-SF Item 3) at baseline	0.126 (0.053)	0.0178	1.134	(1.022,1.258)
ECOG PS grade (0 vs 1)	-0.672 (0.202)	0.0008	0.510	(0.344, 0.758)
Number of bone lesions at baseline (≤10 vs >10)	-0.586 (0.232)	0.0114	0.556	(0.353, 0.876)
Presence of visceral disease (yes vs no)	0.527 (0.216)	0.0146	1.694	(1.110, 2.587)

SE, standard error; CI, confidence interval; PSA, prostate-specific antigen; BPI-SF, Brief Pain Inventory–Short Form; ECOG PS, Eastern Cooperative Oncology Group performance status.

Supplementary Table S4. Relative Dose Intensity in the HRR+ Cohort

	Niraparib+AAP (n=212)	Placebo+AAP (n=211)
Niraparib or placebo RDI (%)		
Median	99.4	100.0
Range	(18.3–115.8)	(29.8–102.6)
Abiraterone RDI (%)		
Median	100.0	100.0
Range	(15.3–146.3)	(14.1–126.4)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; RDI, relative dose intensity.

Supplementary Table S5. Treatment-emergent Adverse Events Leading to Dose Reductions in the HRR+ Cohort

Events, n (%)	Niraparib+AAP (n=212)			Placebo+AAP (n=211)		
	Niraparib	AA	P	Placebo	AA	P
Patients with ≥1 AE leading to dose reduction	42 (19.8)	6 (2.8)	19 (9.0)	7 (3.3)	7 (3.3)	12 (5.7)
Blood and lymphatic system disorders	35 (16.5)	0	0	2 (0.9)	0	0
Anemia	28 (13.2)	0	0	1 (0.5)	0	0
Thrombocytopenia	6 (2.8)	0	0	2 (0.9)	0	0
Neutropenia	3 (1.4)	0	0	0	0	0
Leukopenia	2 (0.9)	0	0	0	0	0
General disorders and administration-site conditions	6 (2.8)	0	3 (1.4)	1 (0.5)	1 (0.5)	1 (0.5)
Fatigue	4 (1.9)	0	1 (0.5)	0	0	0
Asthenia	2 (0.9)	0	0	1 (0.5)	1 (0.5)	0
Face edema	0	0	1 (0.5)	0	0	1 (0.5)
Malaise	0	0	1 (0.5)	0	0	0
Investigations	3 (1.4)	1 (0.5)	0	3 (1.4)	5 (2.4)	0
Blood creatinine increased	2 (0.9)	0	0	0	0	0
Alanine aminotransferase increased	1 (0.5)	1 (0.5)	0	2 (0.9)	4 (1.9)	0
Aspartate aminotransferase increased	1 (0.5)	0	0	2 (0.9)	3 (1.4)	0
Gastrointestinal disorders	2 (0.9)	0	2 (0.9)	1 (0.5)	0	1 (0.5)
Nausea	2 (0.9)	0	0	0	0	0
Constipation	1 (0.5)	0	0	0	0	0
Vomiting	1 (0.5)	0	0	0	0	0

Abdominal pain	0	0	1 (0.5)	1 (0.5)	0	0
Gastric ulcer	0	0	1 (0.5)	0	0	0
Hiatus hernia	0	0	0	0	0	1 (0.5)
Nervous system disorders	2 (0.9)	1 (0.5)	2 (0.9)	0	1 (0.5)	1 (0.5)
Dysgeusia	1 (0.5)	0	0	0	0	0
Lethargy	1 (0.5)	1 (0.5)	1 (0.5)	0	1 (0.5)	0
Dizziness	0	0	1 (0.5)	0	0	0
Headache	0	0	0	0	0	1 (0.5)
Renal and urinary disorders	2 (0.9)	0	0	0	0	0
Chronic kidney disease	1 (0.5)	0	0	0	0	0
Renal impairment	1 (0.5)	0	0	0	0	0
Metabolism and nutrition disorders	1 (0.5)	2 (0.9)	4 (1.9)	0	0	5 (2.4)
Decreased appetite	1 (0.5)	0	0	0	0	0
Diabetes mellitus	0	0	0	0	0	2 (0.9)
Hyperglycemia	0	1 (0.5)	3 (1.4)	0	0	1 (0.5)
Hyperkalemia	0	0	1 (0.5)	0	0	0
Hypokalemia	0	1 (0.5)	0	0	0	0
Type 2 diabetes mellitus	0	0	0	0	0	2 (0.9)
Cardiac disorders	0	2 (0.9)	1 (0.5)	0	0	0
Coronary artery disease	0	1 (0.5)	0	0	0	0
Tachyarrhythmia	0	1 (0.5)	1 (0.5)	0	0	0
Endocrine disorders	0	0	1 (0.5)	0	0	0
Cushingoid	0	0	1 (0.5)	0	0	0
Hepatobiliary disorders	0	0	0	1 (0.5)	0	0
Hyperbilirubinemia	0	0	0	1 (0.5)	0	0

Injury, poisoning, and procedural complications	0	0	3 (1.4)	0	0	1 (0.5)
Contusion	0	0	3 (1.4)	0	0	1 (0.5)
Skin laceration	0	0	1 (0.5)	0	0	0
Musculoskeletal and connective tissue disorders	0	0	1 (0.5)	1 (0.5)	0	1 (0.5)
Muscular weakness	0	0	1 (0.5)	0	0	0
Myalgia	0	0	0	1 (0.5)	0	0
Myopathy	0	0	0	0	0	1 (0.5)
Psychiatric disorders	0	0	2 (0.9)	0	0	1 (0.5)
Insomnia	0	0	1 (0.5)	0	0	1 (0.5)
Personality change	0	0	1 (0.5)	0	0	0
Skin and subcutaneous tissue disorders	0	0	1 (0.5)	0	0	1 (0.5)
Photosensitivity reaction	0	0	1 (0.5)	0	0	0
Sensitive skin	0	0	1 (0.5)	0	0	0
Skin atrophy	0	0	0	0	0	1 (0.5)
Vascular disorders	0	0	1 (0.5)	0	0	1 (0.5)
Hypertension	0	0	1 (0.5)	0	0	0
Lymphedema	0	0	0	0	0	1 (0.5)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; AA, abiraterone acetate; P, prednisone; AE, adverse event.

Supplementary Table S6. Treatment-emergent Adverse Events Leading to Discontinuation in the HRR+ Cohort

Events, n (%)	Niraparib+AAP (n=212)			Placebo+AAP (n=211)		
	Niraparib	AA	P	Placebo	AA	P
Patients with ≥1 AE leading to discontinuation of study agent	23 (10.8)	19 (9.0)	19 (9.0)	10 (4.7)	12 (5.7)	11 (5.2)
Infections	8 (3.8)	8 (3.8)	8 (3.8)	1 (0.5)	1 (0.5)	1 (0.5)
COVID-19	4 (1.9)	4 (1.9)	4 (1.9)	0	0	0
COVID-19 pneumonia	2 (0.9)	2 (0.9)	2 (0.9)	0	0	0
Herpes zoster	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Pneumonia	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Suspected COVID-19	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Blood and lymphatic system disorders	6 (2.8)	2 (0.9)	2 (0.9)	1 (0.5)	1 (0.5)	1 (0.5)
Anemia	5 (2.4)	2 (0.9)	2 (0.9)	1 (0.5)	1 (0.5)	1 (0.5)
Thrombocytopenia	1 (0.5)	0	0	0	0	0
Gastrointestinal disorders	4 (1.9)	3 (1.4)	3 (1.4)	0	0	0
Vomiting	3 (1.4)	2 (0.9)	2 (0.9)	0	0	0
Nausea	2 (0.9)	2 (0.9)	2 (0.9)	0	0	0
Abdominal pain	1 (0.5)	0	0	0	0	0
General disorders and administration-site conditions	4 (1.9)	2 (0.9)	2 (0.9)	1 (0.5)	1 (0.5)	1 (0.5)
Asthenia	3 (1.4)	2 (0.9)	2 (0.9)	0	0	0
Fatigue	1 (0.5)	0	0	0	0	0
General physical health deterioration	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Cardiac disorders	3 (1.4)	3 (1.4)	3 (1.4)	3 (1.4)	3 (1.4)	3 (1.4)

Acute coronary syndrome	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Atrial fibrillation	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Cor pulmonale	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Acute myocardial infarction	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Coronary artery disease	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Myocardial infarction	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Psychiatric disorders	2 (0.9)	2 (0.9)	2 (0.9)	0	0	0
Anxiety	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Completed suicide	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Respiratory, thoracic, and mediastinal disorders	2 (0.9)	2 (0.9)	2 (0.9)	0	0	0
Dyspnea	2 (0.9)	2 (0.9)	2 (0.9)	0	0	0
Musculoskeletal and connective tissue disorders	1 (0.5)	1 (0.5)	1 (0.5)	1 (0.5)	2 (0.9)	2 (0.9)
Arthralgia	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Back pain	0	0	0	0	1 (0.5)	1 (0.5)
Musculoskeletal chest pain	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Reproductive system and breast disorders	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Pelvic pain	1 (0.5)	1 (0.5)	1 (0.5)	0	0	0
Hepatobiliary disorders	0	1 (0.5)	1 (0.5)	0	1 (0.5)	0
Hepatitis acute	0	1 (0.5)	1 (0.5)	0	0	0
Hyperbilirubinemia	0	0	0	0	1 (0.5)	0
Investigations	0	0	0	0	3 (1.4)	2 (0.9)
Alanine aminotransferase increased	0	0	0	0	2 (0.9)	1 (0.5)
Aspartate aminotransferase increased	0	0	0	0	2 (0.9)	1 (0.5)

ECOG PS worsened	0	0	0	0	1 (0.5)	1 (0.5)
Neoplasms benign, malignant, and unspecified (including cysts and polyps)	0	0	0	2 (0.9)	1 (0.5)	1 (0.5)
Acute myeloid leukemia	0	0	0	1 (0.5)	0	0
Lung neoplasm malignant	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Nervous system disorders	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Cerebral arteriosclerosis	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Renal and urinary disorders	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Urinary retention	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Vascular disorders	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)
Circulatory collapse	0	0	0	1 (0.5)	1 (0.5)	1 (0.5)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; AA, abiraterone acetate; P, prednisone; AE, adverse event; ECOG PS, Eastern Cooperative Oncology Group performance status.

Supplementary Table S7. Overall Causes of Death in the HRR+ Cohort

n (%)	Niraparib+AAP (n=212)	Placebo+AAP (n=211)
Patients who died on study treatment[a]	19 (9.0)	19 (9.0)
Adverse event	11 (5.2)	7 (3.3)
Progressive disease	8 (3.8)	12 (5.7)
Patients who died in follow-up[b]	36 (17.0)	40 (19.0)
Adverse event	1 (0.5)	0
Progressive disease	29 (13.7)	31 (14.7)
Other	6 (2.8)	9 (4.3)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone.

[a]On-study treatment death is defined as death that occurs within 30 days of the last dose of study drug.

[b]Follow-up death is defined as the death that occurs >30 days after the last dose of study drug.

Supplementary Table S8. Treatment-emergent Adverse Events Leading to Death in the HRR+ Cohort

Events, n (%)	Niraparib+AAP (n=212)	Placebo+AAP (n=211)
Patients with ≥1 AE leading to death	12 (5.7)	7 (3.3)
Infections	7 (3.3)	2 (0.9)
COVID-19	4 (1.9)	0
COVID-19 pneumonia	2 (0.9)	0
Pneumonia	1 (0.5)	0
Suspected COVID-19	0	2 (0.9)
Cardiac disorders	2 (0.9)	3 (1.4)
Cardiorespiratory arrest	1 (0.5)[a]	0
Cor pulmonale	1 (0.5)	0
Acute myocardial infarction	0	1 (0.5)
Myocardial infarction	0	2 (0.9)
General disorders and administration-site conditions	1 (0.5)	0
Adverse drug reaction	1 (0.5)	0
Psychiatric disorders	1 (0.5)	0
Completed suicide	1 (0.5)	0
Respiratory, thoracic, and mediastinal disorders	1 (0.5)	0
Dyspnea	1 (0.5)	0
Nervous system disorders	0	1 (0.5)
Cerebral arteriosclerosis	0	1 (0.5)
Vascular disorders	0	1 (0.5)
Circulatory collapse	0	1 (0.5)

HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; AE, adverse event; SAE, serious adverse event.

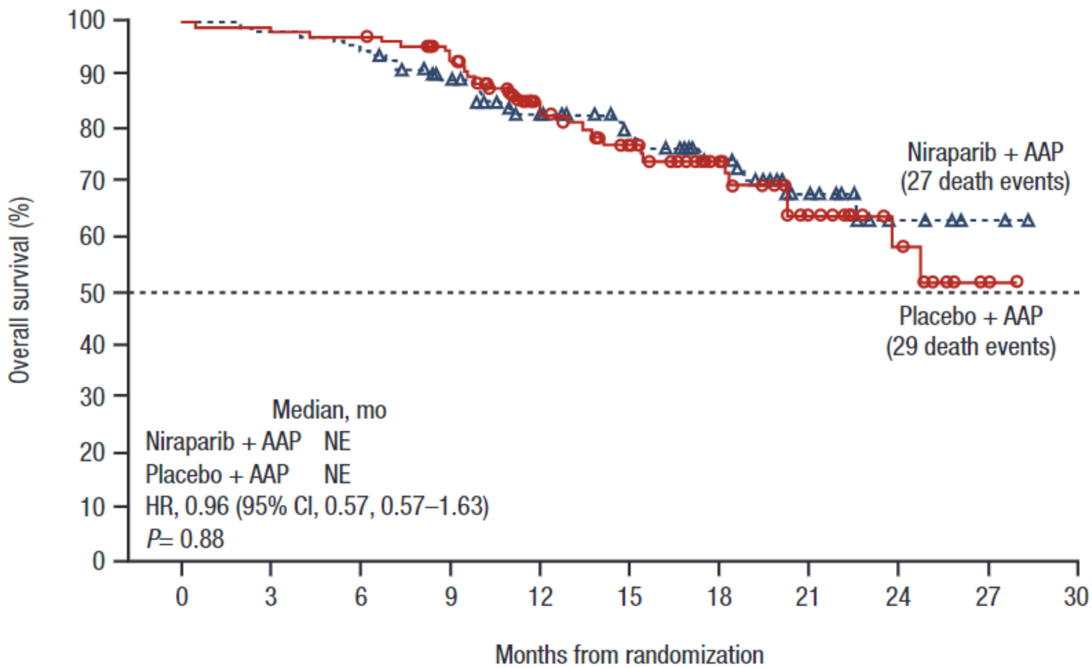
[a]While this patient's underlying cause of death was reported as progressive prostate cancer in Table S7, the proximal SAE term leading to death was reported as cardiopulmonary arrest because disease progression should not be reported as an adverse event.

Supplementary Table S9. Adverse Events in the HRR- Cohort (>10%)

Event, n (%)	Niraparib+AAP (n=123)			Placebo+AAP (n=123)		
	All grades	Grade 3	Grade 4	All grades	Grade 3	Grade 4
Any	121 (98.4)	74 (60.2)	16 (13.0)	120 (97.6)	49 (39.8)	5 (4.1)
Anemia	67 (54.5)	49 (39.8)	1 (0.8)	19 (15.4)	9 (7.3)	0
Hypertension	42 (34.1)	22 (17.9)	0	28 (22.8)	12 (9.8)	0
Constipation	39 (31.7)	3 (2.4)	0	25 (20.3)	1 (0.8)	0
Fatigue	29 (23.6)	8 (6.5)	0	22 (17.9)	6 (4.9)	0
Nausea	29 (23.6)	2 (1.6)	0	20 (16.3)	0	0
Decreased appetite	29 (23.6)	6 (4.9)	0	11 (8.9)	2 (1.6)	0
Thrombocytopenia	25 (20.3)	5 (4.1)	2 (1.6)	3 (2.4)	1 (0.8)	0
Neutropenia	22 (17.9)	9 (7.3)	5 (4.1)	6 (4.9)	2 (1.6)	1 (0.8)
Asthenia	22 (17.9)	7 (5.7)	1 (0.8)	14 (11.4)	0	0
Weight decreased	20 (16.3)	1 (0.8)	0	8 (6.5)	0	0
Vomiting	19 (15.4)	3 (2.4)	0	11 (8.9)	0	0
Back pain	19 (15.4)	2 (1.6)	0	25 (20.3)	0	0
Hyperglycemia	18 (14.6)	2 (1.6)	0	10 (8.1)	0	1 (0.8)
Peripheral edema	17 (13.8)	0	0	15 (12.2)	1 (0.8)	0
Dyspnea	16 (13.0)	4 (3.3)	0	9 (7.3)	1 (0.8)	0
Hypokalemia	15 (12.2)	4 (3.3)	1 (0.8)	9 (7.3)	2 (1.6)	0
Dizziness	13 (10.6)	2 (1.6)	0	13 (10.6)	0	0
Arthralgia	12 (9.8)	0	0	20 (16.3)	0	0

Fall	9 (7.3)	2 (1.6)	0	16 (13.0)	1 (0.8)	0
Hot flush	6 (4.9)	0	0	13 (10.6)	0	0

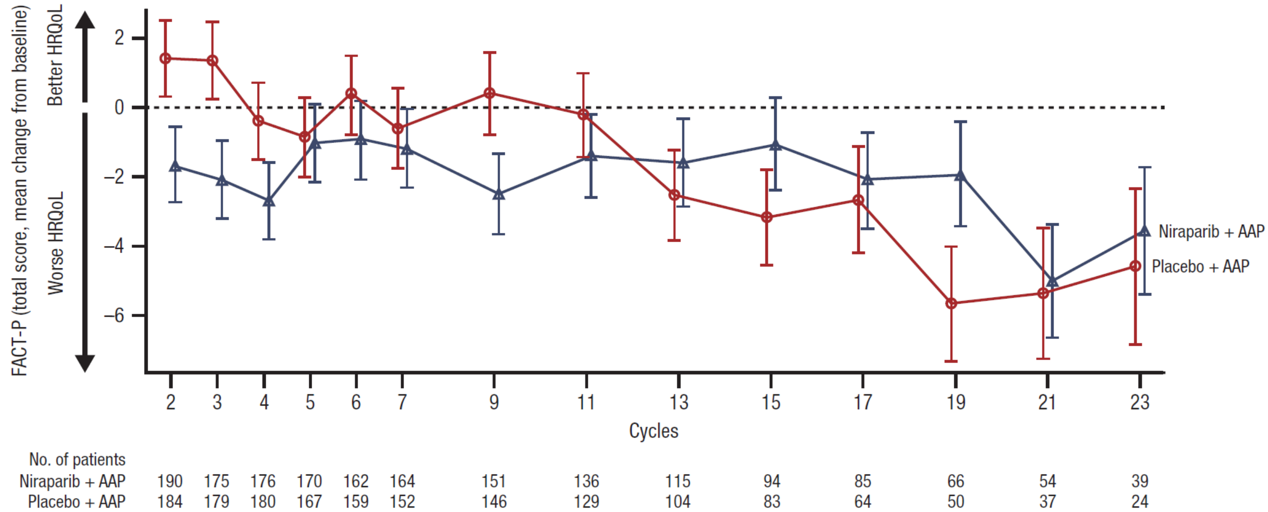
HRR, homologous recombination repair; AAP, abiraterone acetate with prednisone; AE, adverse event.



No. at risk											
Niraparib + AAP	113	111	107	92	65	48	38	21	8	2	0
Placebo + AAP	112	110	109	99	63	52	35	20	10	2	0

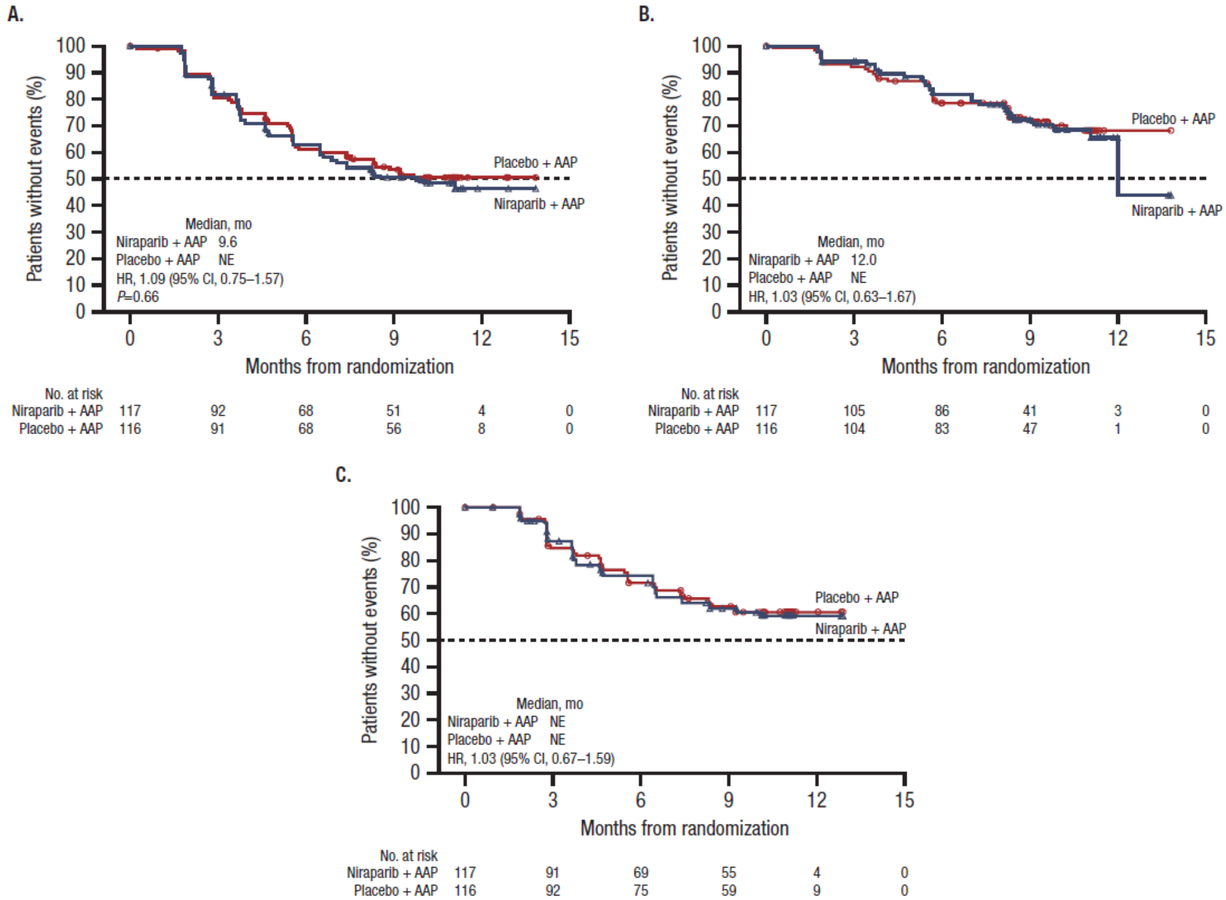
Supplementary Figure S1. Kaplan-Meier Estimate of OS in the *BRCA1/2* Subgroup

OS, overall survival; AAP, abiraterone acetate plus prednisone; NE, not evaluable; HR, hazard ratio; CI, confidence interval.



Supplementary Figure S2. Least-squares Mean Change From Baseline in FACT-P Total Scores in the HRR+ Cohort

FACT-P, Functional Assessment of Cancer Therapy–Prostate; HRR, homologous recombination repair; HRQoL, health-related QoL; AAP, abiraterone acetate plus prednisone.



Supplementary Figure S3. Kaplan-Meier Estimates of the Futility Analyses in the HRR- Cohort.

Kaplan-Meier estimates of composite endpoints in the HRR- cohort (Panel A). Kaplan-Meier estimates of radiographic progression-free survival in the HRR- cohort (Panel B). Kaplan-Meier estimates of prostate-specific antigen progression in the HRR- cohort (Panel C).

HRR, homologous recombination repair; AAP, abiraterone acetate plus prednisone; NE, not evaluable; HR, hazard ratio; CI, confidence interval.

References

1. Scher HI, Morris MJ, Stadler WM, et al: Trial design and objectives for castration-resistant prostate cancer: updated recommendations from the Prostate Cancer Clinical Trials Working Group 3. *J Clin Oncol* 34:1402-1418, 2016
2. Eisenhauer EA, Therasse P, Bogaerts J, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer* 45:228-247, 2009
3. de Bono J, Mateo J, Fizazi K, et al. Olaparib for metastatic castration-resistant prostate cancer. *N Engl J Med* 382:2091-2102, 2020
4. LYNPARZA® (olaparib) [package insert]. Wilmington, DE: AstraZeneca Pharmaceuticals; May 2020